# Split plan: moving the prognostic material and the F-family out of paper 02

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**Status.** Plan only. No manuscript, driver or data file has been changed. Line numbers refer to `report.Rmd` at commit 8e810ca9d (2026-10-29). They will drift as soon as the first cut is made, so work from the section labels.

## Notation

Symbols follow `analysis/report/NOTATION.md`. Code names are in `monospace`. Two symbols have no NOTATION.md entry yet. They are marked *proposed* and should be added when the split is adopted. Paper 02's current text uses retired forms ($c_{pb}$, $c_{tv}$, $c_{bl}$, $D_e$). The cut should rename them to the forms below. Do not declare them as departures.

| Symbol | Meaning | Code |
|---|---|---|
| $B_i$, $B_i^c$ | biomarker of participant $i$; centered at the trial's baseline sample mean | `bm`, `bmc` |
| $w_t$ | calendar week of visit $t$ | `t` |
| $c_{bm}$ | biomarker-response correlation on drug: the interaction under covariance moderation | `c_bm`, `c.bm` |
| $c_{bm}^*$ | ceiling: the largest $c_{bm}$ for which every path's correlation matrix is positive definite | |
| $c_{bm,PB}$, $c_{bm,TV}$, $c_{bm,BL}$ | prognostic correlations: of the biomarker with the placebo-response component, the natural-history component, and the baseline level | `c_pb`, `c_tv`, `c_bl`; `c.bm.pb` |
| $b_{TV}$ (*proposed*) | constant natural-history mean shift per SD of the biomarker; absorbed by the biomarker main effect | `b_tv`, `b.bm.tv` |

| Symbol | Meaning | Code |
|---|---|---|
| $b^{\text{tr}}_{TV}$ (*proposed*) | trend-pathway coefficient: shift of $b^{\text{tr}}_{TV}\,\sigma_{TV}\,z_i\,g(w_t)$, with $z_i$ the standardized biomarker and $g$ the TV mean course scaled to peak 1 | `b_tv_trend`, `b.bm.tv.trend` |
| $c_1$ | cross-component correlation at the same visit (refined process 0.2) | `c.cf1t`, `c.cfct` |
| $t_{1/2}$, $t^a_{1/2}$ | carryover half-life of the DGP; half-life assumed by the analysis | `t1half`; `h`, `hl_analysis` |
| $D_{bc,it}$ | exposure-weighted drug indicator built at $t^a_{1/2}$ | `Dbc` |
| $\eta_t$ | design expectancy: 1 open label, 0.5 blinded, 0 at baseline | `e`; regressor `De` |
| $\beta_\eta$, $\beta_{bm:\eta}$ | coefficients of the expectancy terms $\eta_t$ and $B^c_i\eta_t$ | `De`, `bmc:De` |
| $\kappa$ | mean model-based SE divided by the empirical SD of the estimates | `kappa` |
| $\hat{\pi}$, MCSE | estimated rejection rate; its Monte Carlo SE, $\sqrt{\hat{\pi}(1-\hat{\pi})/n_{\text{reps}}}$ | `reject`, `mcse` |

Paper 02's `bm` is already centered by the DGP (mean 0), so `bm` and `bmc` coincide in its drivers. The remedy term is written `bmc:t` ($B^c_i w_t$) throughout this plan, and its quadratic variant adds $B^c_i w_t^2$.

## Glossary

- **Covariance moderation.** The data-generating convention in which the interaction is a correlation between the biomarker and the drug response, present on drug and decaying off drug. Paper 02 calls it the "correlation encoding"; code labels it `mvn`.
- **Refined process.** The recommended DGP: graded coupling, separable AR(1) with $\rho = 0.7$ and $c_1 = 0.2$, anchored carryover, and a stop on

non-positive-definite matrices. Paper 02 calls it the "extended DGP". **Published process:** Hendrickson's code at commit 58b32a9.

- **Predictive biomarker.** A biomarker that changes the drug response ($c_{bm} \neq 0$). **Prognostic biomarker:** one associated with the outcome through a channel other than the drug response. A biomarker can be both.
- **Prognostic pathway.** One channel of prognostic association:
  - **prog-BL:** $c_{bm,BL}$, baseline level.
  - **prog-TV-const:** $c_{bm,TV}$, natural-history component.
  - **prog-PB:** $c_{bm,PB}$, placebo-response component.
  - **constant shift:** $b_{TV}$, stored as `prog-TV-grow` despite being constant.
  - **trend pathway:** $b_{TV}^{\text{tr}}$, a natural-history mean shift that grows over the trial.

  The first three are correlations and are bounded by feasibility. The last two are mean shifts and are not.
- **Feasibility bound.** The largest prognostic correlation, given $c_{bm}$, for which the joint correlation matrix is positive definite on every path. It has a closed form via the Schur complement (`docs/65`).
- **Silent repair; repair artifact.** Before 2026-10-26 the drivers replaced a non-positive-definite matrix with a nearby positive-definite one without warning. In the strength-0.6 prognostic cells this attenuated the interaction and showed up as a power loss in every analysis (`docs/65`). Repair is now refused by default.
- **G-family.** The analysis ladder of paper 02, stored with `E` codes. The members this plan names:
  - **G1** (`E1`): binary drug indicator, `corCAR1`.
  - **G6** (`E1cr2`): G1 with CR2.
  - **G8** (`E2cr2`): $D_{bc}$ at the true half-life, with CR2.
  - **G9** (`E7cr2`): $D_{bc}$ at a half-life chosen by AIC on the interaction model, with CR2. Slightly liberal (about +0.005); the side analysis after the split.
  - **G9-null** (`E7null_cr2`): half-life chosen by AIC on the model without the interaction, then the interaction model refitted and tested with CR2. **The recommended analysis of paper 02** (Section 2.8). It may be renamed (decision 5).
  - **G8 at a fixed half-life** (`E3fix_cr2`): a prespecified value of 0.25 or 1 week.
  - **Oracle benchmark.** An analysis given the true half-life (G3, G8). It is not available to a trialist. It serves only as the reference that prices the selection step.
- **E0.** Hendrickson's original analysis: `lmer`, random intercept, binary indicator, and the expectancy main effect where expectancy varies. It stays in paper 02.

- **F-family.** The G3 formula plus $\eta_t$ and $B_i^c \eta_t$, with CR2. Members:
  - **F1:** $t_{1/2}^a$ fixed at the 0.2-week policy value.
  - **F1A:** chosen by AIC with the interaction in the model.
  - **F1A0:** chosen by AIC without it.
  - **F1A0B:** F1A0 with a parametric-bootstrap calibration that repeats the selection.

  The family moves to paper 15.
- **Joint recommendation.** The single analysis advice that papers 02 and 15 give together (Section 4).
- **CR2.** The Bell-McCaffrey bias-reduced cluster-robust standard error, used with Satterthwaite degrees of freedom.
- **Scenario family.** A set of simulation cells sharing one departure from paper 02's original grid (Section 2.3).
- **Block (N, A, T, S).** The four blocks of `36-run-referee-variants.R`:
  - **N:** null cells.
  - **A:** alternative cells.
  - **T:** trend pathway.
  - **S:** patient slope variance.
- **Patient slope variance.** A patient-specific linear time trend unrelated to the biomarker (`sd_slope`). It is a heterogeneity stress, not a prognostic one, and stays in paper 02.
- **Two-MCSE band.** The interval of 0.05 plus or minus two MCSE: $[0.036, 0.064]$ at 1,000 replicates.
- **Hybrid, OL+BDC, CO.** The designs: open-label titration then blinded discontinuation then a brief crossover; open-label titration then blinded discontinuation; two-period crossover.

## 1. Decision

Paper 02 is narrowed to one question: how to represent carryover and choose the analysis half-life for the biomarker-treatment interaction when the biomarker is predictive only. The data come from the refined process under covariance moderation, meaning the biomarker-response correlation is the interaction mechanism.

Everything about a biomarker that is also prognostic moves to a new companion paper, provisionally paper 15. So does the F-family (F1, F1A, F1A0, F1A0B), whose expectancy terms exist to protect against a prognostic biomarker.

**Recommended analysis (decided 2026-10-30).** Paper 02 recommends **G9-null**:

- $D_{bc}$, with the analysis half-life chosen by AIC over $\{0.25, 0.5, 1, 2\}$ weeks on the model *without* the interaction

- the interaction model refitted by REML at that half-life
- the test of `bm:Dbc` by CR2 with Satterthwaite df

G9, which selects on the model *with* the interaction, becomes the comparator that shows what that choice costs. The reason is size. In the paired block N evidence, G9 is liberal against the oracle in every cell, while G9-null cannot be distinguished from it (Section 2.8). G9-null gives up 1 to 3 points of power for that. Among analyses that do not need the true half-life, it is the most powerful one that holds nominal size. That is the sense in which it is "the best model" in paper 02. It is not the most powerful analysis, and the paper must say so. Section 2.8 lists what has to be run before the paper can carry this claim.

A note on wording: the narrowed paper 02 does not use the published process at 58b32a9. Step coupling in that process removes the signal at any positive half-life (`docs/55`). Describe the scope as "the refined process under covariance moderation, with a predictive-only biomarker".

**Rationale.**

- The split removes two of paper 02's submission blockers outright:
  - the repaired strength-0.6 rows
  - the untested `bmc:t` remedy

  Both become properly scoped results in paper 15, where the 0.6 cells are the repair-artifact exhibit and the `bmc:t` check becomes a tested analysis.
- It cuts about 15 to 20% of the manuscript.
- It gives the prognostic material a coherent paper of its own. That paper is largely drafted already across `docs/47`, `docs/49`, `docs/56`, `docs/64`, `docs/65` and `docs/series/ch19`.

## 1a. Decisions needed before work starts

Five points emerged while mapping the split. Decision 1 is resolved. The others need an author decision; the detail is in the section cited.

1. **Which F-family member carries the joint recommendation (Section 4). Resolved 2026-10-30.** With G9-null as paper 02's analysis, the expectancy-term counterpart is F1A0. That is G9-null's selection rule plus $\eta_t$ and $B_i^c\eta_t$, in its bootstrap-calibrated form F1A0B. This agrees with CLAUDE.md's provisional recommendation.

   F1, which fixes the half-life at 0.2 week, is not the recommendation. In paper 02's cells it loses up to 10 points of power to a misspecified half-life.

2. **Large-N size evidence (Section 2.6).** Family 6, the $N \in \{140, 280\}$ cells, is entirely prog-PB. After the split, paper 02 has no G9 size evidence above $N = 70$. There are two options:

- **(a)** State the $N \leq 70$ limit, which the paper already partly does.
- **(b)** Run a confound-free large-N null. The cost is a rough estimate of 1 to 3 hours on zeus, not a timing.

3. **Overlap between paper 01 Study 3 and paper 15 (Section 5.2).** Paper 01 already contains a prognostic stress test and a joint feasibility table (`tab:prognostic`). The recommendation is that paper 15 cite paper 01 for the DGP-level facts and present only the analysis-level consequences. Paper 01, which is near submission at grade B, would not be reopened.

4. **Role of the models that fix the analysis half-life.** The author's position (2026-10-30) is that models assuming a known half-life lack face validity. Three groups are affected:

   - **(i) Oracle analyses at the true half-life (G3, G8).** Recommended: keep, labeled "oracle benchmark, not a candidate analysis" throughout. G8 is the reference that prices the selection step, so without it the size evidence of Section 2.8 cannot be stated.
   - **(ii) Analyses at a prespecified value (G8 at 0.25 or 1 week).** Recommended: keep as evidence of what a wrong guess costs, but never as a candidate. F1 leaves with the F-family.
   - **(iii) Binary-indicator analyses (E0, G1, G6).** These fix the half-life at 0. They stay as the published and default reference analyses.

   Alternative: drop group (ii) entirely. Decision pending.

5. **Name for G9-null.** It is now the recommended analysis, so a name with the suffix "-null" undersells it. There are two options:

   - **(a)** Keep "G9-null" (stored as `E7null_cr2`) and explain it.
   - **(b)** Give it a ladder code, for example G10, and record that in NOTATION.md.

   Recommended: (b), with G9 described as "G10 selecting on the interaction model". Decision pending.

## 2. Paper 02 after the split

### 2.1 Scope statement (Introduction and Methods)

Add one paragraph to the Introduction. Suggested text:

> The biomarker in this study is predictive only: it changes the drug response and nothing else. A biomarker that is also prognostic raises a different question, namely whether the interaction test confuses

an association with the natural history or the placebo response with moderation of the drug effect. A constant prognostic shift is absorbed exactly by the biomarker main effect of every analysis here and cannot move the interaction estimate. Prognostic associations that vary over time, and the design-expectancy terms that protect against them, are the subject of a companion paper [paper 15].

Keep the absorption check for the constant shift ($b_{TV}$, "prog-TV-grow" in the stored data). It takes one sentence and answers the first question a referee will ask.

### 2.2 Section-by-section cut list

Status values:

- **Delete:** remove the section.
- **Trim:** remove the prognostic or F1 content only.
- **Rewrite:** recast the text without it.
- **Keep:** leave as it is.
- **Move:** relocate the material to paper 15.

| Section (label) | Lines | Action |
|---|---|---|
| Setup chunk: `s28`, `s30`, `pdc`, `ceil` loads | 122-129 | Delete. These are the prognostic stress, random-effects, PD-check and ceiling data. |
| Setup chunk: `m_lab`, `focus`, `focus_cols`, `rv_lab` | 140-164 | Trim. Remove F1, F1A, F1A0 and F1A0B. |
| Setup chunk: `rvT` | 168 | Delete. Keep `rvS`; see the slope row below. |

| Section (label) | Lines | Action |
|---|---|---|
| Abstract | 230-273 | Rewrite: (a) Methods: drop “or add expectancy terms (F1, F1A0B)”. (b) Results: drop the F1 sentence, the trend-bias sentence and the strength-0.6 artifact sentence. (c) “Eight scenario families” becomes six; see 2.3. (d) Conclusions: recommend G9-null, not G9, and add “for a biomarker that is predictive only”. (e) Methods: rung 4 is G9-null; G9 is the side analysis that selects on the interaction model. (f) Results: report the G9-null size against the oracle, and G9’s half-point excess as the cost of selecting on the tested model. |
| Introduction, ladder overview | 385-400 | Trim. Remove the F-family sentence and the prognostic, trend and large-N items from the list of stresses. Add the scope paragraph from 2.1. |
| Aims (`sec-aims`) | 457-480 | Trim. Remove “prognostic pathways, large samples, a biomarker that predicts the natural-history trend” from the second aim. Keep “patient-specific slope variance”. |

| Section (label) | Lines | Action |
|---|---|---|
| Table `tab-families` | 482-500 | Rewrite. Delete rows 5 (prognostic) and 6 (large N; every cell there is prog-PB). Row 7 becomes slope variance only. Renumber. |
| `sec-ladder-def`, table `tab-ladder-def` | 643-672 | Rewrite. Remove the pointers to the F-family. Redefine rung 4 as G9-null (half-life chosen by AIC without the interaction). Label G3 and G8 as the oracle benchmark (decision 4). |
| `sec-g4` (rung 4) | 769-802 | Rewrite. Lead with null-model selection and why it keeps the test honest. Then present selection on the interaction model (G4, G9) as the variant whose size cost is measured. The existing F1A/F1A0 explanation (793-794) carries over. |
| `sec-e0` | 673-688 | Keep. E0 contains the expectancy regressor `De` because Hendrickson's analysis does. That is part of the original analysis, not an F-family feature. Add one clause noting `De` is constant, and so dropped, under CO. |

| Section (label) | Lines | Action |
|---|---|---|
| `sec-comparators` | 803-839 | Rewrite. Delete the F-family paragraph (807-831). G9-null moves out of the side analyses into rung 4. Side analyses become G9 (selection on the interaction model) and G8 at a prespecified half-life. Retitle "Side analyses: variants of the selection". |
| `sec-pathways` | 841-866 | Delete, except one sentence on constant-shift absorption, which goes in the scope paragraph. |
| `sec-choice` | 868-886 | Keep. The earlier `bm:L` and `bm:tsd` models are a carryover question. |
| `sec-grid` | 924-960 | Trim. Remove the prognostic-cell clauses (924, 929, 950-955). |
| `sec-corrections` | 1008-1058 | Trim. The PD-refusal correction stays as a statement about the driver. Its prognostic-scenario evidence moves to 15 (1041-1044). |
| Ladder-at-a-glance (`sec-glance`) and Fig. 1-3 captions | 1072-1274 | Rewrite after the ladder script is rerun (2.4). Remove F1 and F1A0B from the captions (1127, 1159) and from the prose (1206-1207, 1271). |
| `sec-rung1`, E0 isolation | 1282-1580 | Keep. The expectancy mentions here describe E0's `De` term, which stays. |
| Rung 4 null variants (`stage3-variants-null`) | 2077-2125 | Trim. Remove the F1 column. |

| Section (label) | Lines | Action |
|---|---|---|
| `sec-alternatives` | 2172-2341 | Rewrite. Keep the fixed-half-life and null-model-selection alternatives. Delete the F-family evidence (2176-2183, 2279-2339, the 'G9>F1' paired step). Move the G9-versus-F1A0B comparison to 15. |
| `sec-scen-grids` | 2353-2457 | Trim. Remove the F1 columns (2381-2394, 2436-2448). |
| `sec-dropout` | 2458-2585 | Trim. Remove the F1 columns (2472-2483, 2499-2500, 2514, 2524-2531, 2565-2566). Line 2533 cites the no-dropout reference of the prognostic stress job. Re-anchor it on a non-prognostic reference or drop the comparison. |
| `sec-prognostic` (all subsections, including `sec-feasibility`, `sec-feasible-grid` and "Larger samples") | 2586-2939 | Move to 15. |
| `sec-trend`: trend pathway | 2940-3012 | Move to 15 (table `stage5-trend`, Fig. 4). |
| `sec-trend`: patient slope variance | 3014-3045 | Keep as its own short subsection, "Patient slope variance", with G8 and G9 only. It is a heterogeneity stress, not a prognostic one: the slope is unrelated to the biomarker. Delete the fig4 trend contrast (3028-3040). |

| Section (label) | Lines | Action |
|---|---|---|
| `sec-random` | 3046-3090 | Move to 15. It was fitted on the prognostic stress cells, 8 of 12 of them the repaired strength-0.6 cells, and was motivated by the artifact. |
| `sec-calibration` | 3091-3178 | Trim. Remove the F1 $\kappa$ values (3142-3143). |
| `sec-which`, ladder score narrative | 3179-3262 | Rewrite from the regenerated table B, with G9-null as the reference analysis ("percent of G9-null power"). "G9 has the largest cell size … under the trend pathway" no longer applies. Remove the F1 power-loss sentence (3250). "For a Hybrid trial G9 is the analysis to use" becomes G9-null, stated together with its power cost against G9. |
| Discussion, `sec-bought` | 3265-3364 | Trim. Remove the F-family and prognostic summaries (3331-3359). |
| `sec-recommend` | 3365-3414 | Rewrite. Recommend G9-null. Give the trade in one sentence: half a point of size bought back for 1 to 3 points of power. State the scope condition (predictive-only biomarker). Insert the joint-recommendation sentence (Section 4). Remove the trend caveat (3404-3406); it moves to 15. |

| Section (label) | Lines | Action |
|---|---|---|
| "What the expectancy terms are for" | 3415-3442 | Move to 15. In 02, replace it with the one-sentence expectancy note in 2.5. |
| Limitations | 3443-3497 | Trim. Remove the repaired-cell, prognostic-grid and large-N F-family sentences (3469-3488). Add the large-N gap (2.6). |
| Conclusions | 3534-3595 | Rewrite items 4, 5, 6, 7 and 8. Item 4 becomes "choose the half-life on the model without the interaction". Item 7 recommends G9-null. Item 5 keeps the prespecified-half-life and interaction-model-selection findings. Item 6 (feasibility) moves to 15. Item 8 loses the prog-PB and trend sentences and gains the scope condition. |
| Glossary | 3596-3715 | Trim. Remove F1, F1A, F1A0, F1A0B, the prognostic labels and the feasible-grid note (3598, 3625, 3642). |
| Reproducibility | 3751-3878 | Trim. Remove the jobs 28 and 30, scripts 30 and 31, and block T entries (3786-3792, 3829). |

### 2.3 Scenario families after the split

| New | Old | Content |
|---|---|---|
| 1 | 1 | Original 30-cell grid |

| New | Old | Content |
|---|---|---|
| 2 | 2 | Misspecified half-life and decay shape |
| 3 | 3 | Serial correlation |
| 4 | 4 | Patient-level dropout |
| 5 | 7 (block S only) | Patient slope variance |
| S | S (block N: $c_{bm,PB} = 0$ cells only; block A) | Independent-seed Hybrid run |

Block N of the referee variants holds four null cells at $c_{bm,PB} = 0.3$. They leave family S. The pooled G9 null figure in the abstract ("7,987 null fits … 0.055") must be recomputed without them. Check whether `m2-pooled-size.csv` (script 34) already pools only $c_{bm,PB} = 0$; the ladder script sets `pooled_null = c_pb == 0` for block N, but script 34 has not been checked.

### 2.4 Ladder script (`37-ladder-tables-figures.R`)

The script reads stored data only, so regenerating it is cheap. The changes are:

1. Drop family 5 (job 28 and the gap-fill sub `C1-ladder`), family 6 (sub `C2-largeN`), block T, and the block N cells with $c_{bm,PB} > 0$.
2. Drop F1 and F1A0B from `analysis_cols`, the paired steps (the 'G9>F1' row) and `side_x`. Drop them from the Fig. 2 and Fig. 3 analysis lists.
3. Make G9-null rung 4 and the reference analysis:
   - Change `mean_power_pct_of_g9` to percent of G9-null.
   - Set the paired steps to G8 to G9-null (rung 3 to 4) and G9-null to G9 (side).
   - Move G9 to `side_x`.

   This needs the backfill of Section 2.8 first. Until then G9-null exists in family S only.
4. Delete Fig. 4 (trend). Paper 15 will need its own version.
5. Rerun the script and check that tables A, B and C and Fig. 1-3 still satisfy the Rmd's `stopifnot` guards.
6. Re-read every inline `lv()`, `inc()` and `bv()` call against the new tables.

Rather than deleting code, add a `--scope predictive` switch that filters these out. Paper 15 can then run the same script with `--scope all` for its G-versus-F comparisons.

### 2.5 Expectancy note for 02

A referee will ask why the G-family omits a design covariate that varies in Hybrid and OL+BDC. Answer in one sentence in `sec-recommend`:

> With a predictive-only biomarker the G-family holds nominal size without the design-expectancy terms in every family examined; the terms matter when the biomarker may also be prognostic, which the companion paper [15] examines.

Do not claim more than that. The claim that F1 matched G9's power at every alternative belongs to paper 15, which carries the evidence.

### 2.6 Evidence gap the split opens

Family 6 (large N) consists entirely of prog-PB cells. After the split, paper 02 has no G9 size evidence above $N = 70$ in the predictive-only case. Choose one of two options:

- **(a) State the limit.** Size is verified to $N = 70$. This matches what the paper already says ("measured only up to $N = 70$ with complete data").
- **(b) Run a confound-free large-N null.** Hybrid, $N \in \{140, 280\}$, $c_{bm} = 0$, $c_{bm,PB} = 0$, G8, G9-null and G9, 1,000 replicates, through `32-run-g9-gap-fill.R` or a new block of `36-run-referee-variants.R`. The section the cut removes already asked for this check ("a targeted size check at larger $N$ with a confound-free counterpart"). Cost is not timed. G9 refits five models per replicate, and at $N = 280$ each fit is several times slower than at $N = 70$. A rough estimate is 1 to 3 hours on zeus. Time it with `--time` first.

Option (a) is sufficient for submission. Option (b) is cheap and makes the paper stronger.

### 2.7 Effort for 02

- 1 to 2 days of rewriting.
- About 1 hour for the ladder script and checks.
- The G9-null backfill (2.8):
  - half a day for the driver
  - an estimated 2 to 5 hours on zeus (untimed)
- Optionally the large-N null (2.6(b)), which now includes G9-null.

### 2.8 G9-null as the recommended analysis: evidence and backfill

**Evidence in hand.** Block N, job 34: Hybrid, 5,000 replicates per null cell, with G8, G9 and G9-null fitted to the same trials. The table shows the four cells with $c_{bm,PB} = 0$, the ones paper 02 keeps.

| Cell | G8 (oracle) | G9 | G9-null | G9 − G8 (paired SE) | G9-null − G8 (paired SE) |
|---|---|---|---|---|---|
| $t_{1/2} = 0.5$, $N = 35$ | 0.048 | 0.052 | 0.047 | +0.004 (0.0016) | \$-\$0.001 (0.0015) |
| $t_{1/2} = 0.5$, $N = 70$ | 0.051 | 0.054 | 0.051 | +0.002 (0.0014) | \$-\$0.000 (0.0014) |
| $t_{1/2} = 1$, $N = 35$ | 0.051 | 0.060 | 0.051 | +0.009 (0.0019) | \$-\$0.000 (0.0017) |
| $t_{1/2} = 1$, $N = 70$ | 0.046 | 0.051 | 0.045 | +0.004 (0.0017) | \$-\$0.001 (0.0016) |

The two-MCSE band at 5,000 replicates is $[0.044, 0.056]$. The results are:

- **G9 is liberal against the oracle in every cell.** It is in all 8 block N cells, including the prognostic ones, and 7 of those 8 are above 2 paired SE.
- **G9-null is within 1 SE of the oracle everywhere.**
- **The standard errors agree with this.** $\kappa$ is 0.89 to 0.97 for G9 and 0.94 to 0.99 for G9-null.
- **Power.** G9-null's cost is 1 to 3 points, from block A: 8 alternative cells at 1,000 replicates, Hybrid only. In ladder table B it reaches 96.6% of G9's power.

**What is missing.** G9-null was fitted only in family S (blocks N and A, Hybrid, $N \leq 70$). A recommended analysis must be shown in every family the paper reports:

- family 1, the 30-cell grid over CO, Hybrid and OL+BDC (job 27)
- family 2, misspecified half-life and decay shape (job 21 cells and gap-fill B1, B2)
- family 3, serial correlation (gap-fill B4)
- family 4, patient-level dropout (gap-fill A1, plus the A2 size diagnostic)
- the slope-variance cells (block S)

**Backfill run.** Write a new driver, `40-run-g9-null-backfill.R`:

- It loops over the cell grids of `27-run-tier1-g-vs-f-e0.R`, `32-run-g9-gap-fill.R` (sub-blocks A1, A2, B1, B2, B4), `06-g-vs-f-comparison.R` (the four Stage 4 cells) and block S of `36-run-referee-variants.R`.
- For each (cell, replicate) it fits `E7null_cr2` and `E7cr2`. `fit_spec_s7()` already implements both.

- **Pairing.** Each driver seeds a replicate as `set.seed(base + r)` before generating data, so rerunning on zeus with the same seeds regenerates the same trials. G9 is refitted only as a check: its estimate must match the stored value to $10^{-8}$ in every replicate.
- **If the check fails** (a DGP or image change since the original job), the stored G9 cannot be paired. Rerun G8, G9 and G9-null together in that sub-block and pair within the new run.
- **Size:** roughly 55,000 replicates.
- **Cost:** about two G9-sized fits per replicate. Estimated at 2 to 5 hours on zeus's 8 cores, but untimed; run `--time` first.

**What would overturn the recommendation.** Two outcomes would:

- a family in which G9-null leaves the size band while G9 does not
- a family in which G9-null loses much more than 3 points of power

The CO design deserves attention. There the candidate half-lives differ least in their fixed-effect fit, so null-model AIC carries little information about the half-life.

Expected grade after the cut: B, with the front matter (cover letter, title page, declarations) still outstanding as in `docs/66`.

## 3. Paper 15 outline: prognostic biomarkers and the interaction test

Working title: *When the biomarker is also prognostic: bias in the biomarker-treatment interaction test of aggregated N-of-1 trials and two remedies.* Directory: `analysis/report/15-prognostic-biomarker/`. Target: the same venue as 01 and 02, as a third companion, or a methods journal on its own.

| Section | Content | Source (existing) |
|---|---|---|
| 1. Introduction | Predictive versus prognostic biomarkers; why the interaction test can confuse the two in N-of-1 designs with varying expectancy | `docs/49`, `ch19`, lexicon `docs/29` |

| Section | Content | Source (existing) |
|---|---|---|
| 2. Taxonomy of pathways | Constant shift (absorbed); correlation pathways prog-BL, prog-TV-const and prog-PB; mean-shift trend pathway (unbounded) | 02 `sec-pathways`; `tidyverse/R/functions.R` trend block |
| 3. Feasibility | Closed-form joint bound on the prognostic correlation and $c_{bm}$ (Schur complement); feasible grid | `docs/65`; scripts 30 and 31; 02 `sec-feasibility`. **Paper 01 already has a joint-ceiling table** (`tab:prognostic`, `18-prognostic-joint-ceiling.R`). Cite 01 for it rather than duplicating; see 5.2. |
| 4. The repair artifact | Strength-0.6 cells: power loss from silent PD repair, not confounding | `docs/65`; jobs 28 and 30; 02 `sec-prognostic` "strength-0.6" subsection; `sec-random` |
| 5. Analyses | G-family (G8, G9) against the F-family (F1, F1A, F1A0, F1A0B); expectancy terms $\eta_t$, $B_i^c\eta_t$ (regressors `De`, `bmc:De`); why they absorb the expectancy-scaled on/off contrast under prog-PB | `docs/56`, `docs/57`, `docs/62`, `docs/64`; 02 `sec-comparators` and "What the expectancy terms are for" |
| 6. Correlation pathways: results | Size and power at strength 0.3 and on the feasible grid; large N under prog-PB | Job 28; gap-fill `C1-ladder` and `C2-largeN`; 02 tables `stage5-*` |
| 7. Trend pathway: results and the `bmc:t` remedy | Bias of every analysis; linear and quadratic `bmc:t`; cost when there is no trend | Block T (job 36); **new run** (3.1) |

| Section | Content | Source (existing) |
|---|---|---|
| 8. G-family against F-family without a prognostic effect | The F-family's cost when unneeded; F1's fixed-half-life penalty versus F1A0B | Job 27 (`tier1-g-vs-f-e0`), job 21 (`g-vs-f-comparison`); 02 `sec-alternatives` F-material |
| 9. Recommendation | Joint recommendation (Section 4) | |
| Appendix | Constant-shift absorption proof; repair mechanics | `docs/65` |

### 3.1 New simulation for paper 15

**`bmc:t` block.**

- Add specifications `E2cr2_bmt`, `E7cr2_bmt` and `F1_bmt` to `simulation-core.R`: the existing formula plus `bmc:t`. For G9, add the term to both the AIC-selection formula and the refit formula. Optionally add quadratic variants.
- Add a new block to `36-run-referee-variants.R`. Cross the trend pathway $b_{TV}^{\text{tr}} \in \{0, 0.3\}$ with $c_{bm} \in \{0, 0.45\}$ at Hybrid, $N = 70$, $t_{1/2} = 0.5$, 1,000 replicates, plus a 5,000-replicate null trend cell for size.
- Estimate about 30 to 45 minutes on zeus. Block T took 8 minutes for 2 cells and 3 specifications.

A local pilot on 2026-10-29 (scratch script, 300 replicates, G8 only) gave the following:

- The null bias of +0.26 fell to +0.10 with linear `bmc:t` and to +0.00 with quadratic.
- Power fell from 0.96 to 0.94 and 0.92.

The DGP's trend follows the TV Gompertz course, not a straight line, so linear `bmc:t` is expected to be only a partial remedy.

### 3.2 Effort for paper 15

About 1 to 2 weeks:

- Drafting is mostly assembly from the sources above.

- One new simulation block (about 1 hour of compute plus half a day of code).
- The ladder script runs with `--scope all` for the G-versus-F figures.

## 4. Joint recommendation (both papers, CLAUDE.md, NOTATION.md)

The two papers must give one answer between them. Present the F-family as the G-family plus $\eta_t$ and $B_i^c\eta_t$ (regressors `De` and `bmc:De`):

- **Paper 02** settles the carryover layer with G9-null. That means an exposure-weighted predictor, with the half-life chosen by AIC on the model without the interaction, and CR2 with Satterthwaite df.
- **Paper 15** adds the expectancy layer when the biomarker may be prognostic, and a `bmc:t` layer if the biomarker may predict the natural-history trend.

**Resolved 2026-10-30: the F-family member is F1A0, and its calibrated form is F1A0B.** F1A0 is G9-null's selection rule with $\eta_t$ and $B_i^c\eta_t$ added. Choosing it makes the two layers nest exactly: G9-null and F1A0 differ only in the expectancy terms. It also matches CLAUDE.md's provisional F1A0B.

Two things remain for paper 15 to show:

- whether F1A0B's bootstrap calibration is needed once the selection excludes the interaction (`docs/64` found no Type I inflation)
- what F1A0 costs against G9-null when the biomarker is not prognostic

F1 at the 0.2-week policy value is not the recommendation. It loses up to 10 points of power where that value is wrong (family 1).

Recommended wording:

> For a biomarker believed to be predictive only, use G9-null: the exposure-weighted predictor with the half-life chosen by AIC on the model without the interaction, tested with CR2 (paper 02). Where the biomarker may also be prognostic, add the design-expectancy terms $\eta_t$ and $B_i^c\eta_t$ (F1A0, paper 15). Where it may predict the rate of natural-history change, also add a biomarker-by-time term.

Once decided, update:

- the CLAUDE.md "Recommended analysis for the interaction (F1)" paragraph
- NOTATION.md (analysis IDs; the new paper number)
- `docs/66` (portfolio table)
- `docs/60` Part 2 (paper 02 plan)

## 5. Cross-paper effects

### 5.1 Paper 01

Paper 01 cites paper 02. Check that none of its citations points at material that moves. Grep for "companion" in 01. As of 8e810ca9d, 01's own prognostic content is its Study 3 stress test, which is independent of 02.

### 5.2 Overlap between paper 01 Study 3 and paper 15

Paper 01 already contains:

- a stress test under a prognostic biomarker (Study 3, `tab:prognostic`)
- a joint feasibility table (`18-prognostic-joint-ceiling.R`)
- the statement that the stop-on-non-PD default came from "a prognostic scenario" with correlation 0.6 (lines 612-618)

Paper 15 must cite 01 for the DGP-level facts: the bound exists, and repair is refused. It should present only the analysis-level consequences. Paper 01 is at grade B and should not be reopened for this.

### 5.3 Portfolio count

The split adds a fifteenth manuscript. `docs/66` recommends consolidating toward about nine. The split is justified because the boundary is scientific, not editorial. It should be recorded alongside the merges already proposed there: 13 with 14, 08 into 10, 12 into 01, and 11 into 01 or 03.

## 6. Order of work

1. Settle decisions 4 (fixed-half-life models) and 5 (G9-null's name), and 2.6 (a) or (b).
2. Write and time `40-run-g9-null-backfill.R` (2.8). Run it on zeus, together with the large-N null if 2.6(b) was chosen. Check the G9 pairing guard before using any result.
3. Confirm that G9-null holds the size band in every family. If a family overturns the recommendation (2.8), stop and revisit before rewriting.
4. Add `--scope` and the G9-null reference to the ladder script, then regenerate the predictive-only tables and figures.
5. Cut and rewrite paper 02 per 2.2. Re-verify every inline number. Run `perl tools/notation-lint.pl`. Render through `tools/render.sh`.
6. Create `analysis/report/15-prognostic-biomarker/` and move the excised Rmd sections into a first draft, keeping their chunks working against the same data.

7. Implement and run the `bmc:t` block (3.1). Include F1A0 alongside G9-null.
8. Update CLAUDE.md (the recommended-analysis paragraph: G9-null, with F1A0/F1A0B as the prognostic-safe form), NOTATION.md, `docs/60` and `docs/66`.

## 7. Unverified assumptions in this plan

- The line ranges come from a grep of section headers and F1 and prognostic mentions, not a line-by-line read. Some cross-references outside the listed ranges will also need cutting.
- I have not checked whether `m2-pooled-size.csv` excludes the $c_{bm,PB} = 0.3$ null cells (2.3).
- The large-N run cost in 2.6(b) is a rough estimate, not a timing.
- The cost of the `bmc:t` block is scaled from block T's runtime. Its findings rest on a 300-replicate G8-only local pilot.
- G9-null's standing as the best-sized analysis rests on the 8 block N null cells and 8 block A alternative cells (Hybrid, $N \leq 70$). Its behavior in CO, OL+BDC, dropout, misspecified-shape and serial-correlation cells is unknown until the backfill runs.
- The backfill's pairing assumes that rerunning the stored seeds on zeus regenerates identical trials. Any DGP or image change since the original jobs breaks this. The G9 refit check (2.8) detects it; nothing has verified it yet.
- The backfill cost (2 to 5 hours) is an estimate, not a timing.